

Target Labels Are Not Portfolio Returns: A Triple-Barrier Failure Mode in Factor Investing

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Abstract

This paper studies a practical failure mode in financial machine learning: treating target-label prediction as sufficient evidence for portfolio performance. Using a proprietary U.S. equity research panel and the GC004 research artifacts from the FactorInvest environment, I evaluate triple-barrier model labels in a factor-investing setting. The primary target, TBM_LONG_FIX_21D_SL5_TP10, defines a long event over a 21-trading-day horizon with a 5% stop-loss barrier and a 10% take-profit barrier. A weighted multiclass ridge classifier produces a positive top-decile take-profit lift of 5.01 percentage points under the weekly sampling policy. However, the same top decile has a negative event-return lift of -0.67 percentage points and a stop-label share of 67.26%.

When the frozen candidate is converted into a temporal-safe mark-to-market portfolio rule and compared with a high-liquidity equal-weight benchmark, the apparent label signal fails economically. At a 50 bps round-trip cost, the portfolio returns -37.90% in the test window versus -13.83% for the benchmark, producing an excess total return of -30.59%. The result remains negative even at zero transaction cost. Subsequent repair attempts, including liquidity filters, stop-probability gates, healthcare exclusion, sector caps, and ranking reformulations, reduce some damage but do not produce a validation-locked rule.

The contribution is a methodological case study rather than a new trading strategy. The central lesson is that triple-barrier labels should be bridged to explicit economic return contracts before being treated as investable signals. A model can learn the label and still fail the trade.

Keywords: financial machine learning; triple-barrier labeling; factor investing; target engineering; backtesting; portfolio validation; label mismatch; research governance

JEL Codes: G11; G12; G17; C53; C58

Disclosure and scope. This paper is an independent research study. It is based on historical simulations using a proprietary U.S. equity research panel. Reported returns are research returns and do not fully include transaction costs beyond the stated sensitivity tests, borrow costs, market impact, taxes, slippage, or all implementation constraints. The paper does not present a live trading strategy, investment recommendation, or production-ready portfolio.

1. Introduction

Financial machine learning pipelines often separate research into two stages. The first stage defines a target label that seems economically meaningful. The second stage trains a model to predict that label. Triple-barrier labeling is attractive because it turns a future price path into an event: an asset may hit a take-profit barrier, hit a stop-loss barrier, or expire at a vertical time barrier. Relative to simple forward-return regression, this can encode path dependence, downside asymmetry, and holding-period constraints.

The danger is that the label is not the portfolio objective. A model may rank assets by the probability of hitting a take-profit barrier and still select names with poor benchmark-relative mark-to-market returns. The selected assets may have higher stop incidence, unfavorable payoff asymmetry, hidden sector concentration, weak liquidity, or simply a poor relationship between event frequency and economic payoff. In such a case, classification success is not portfolio success.

This paper documents one such failure. The research path is deliberately practical. It starts with label evidence, moves through event-return diagnostics, freezes a temporal-safe candidate, evaluates it as a mark-to-market portfolio, and then records repair attempts. The final decision is negative: the rule is not promoted. The value of the paper is precisely that the failure is observed, documented, and converted into a validation lesson.

- First, it documents a full label-to-portfolio failure path in a factor-investing pipeline.
- Second, it shows that positive triple-barrier take-profit lift can coexist with negative event-return lift and negative mark-to-market benchmark-relative performance.
- Third, it proposes a practical validation discipline: event labels should be tested against explicit economic return contracts before they are treated as investable signals.

The central sentence is simple: the model learned the label, but the label was not the portfolio objective.

2. Related Literature and Positioning

This paper sits at the intersection of financial machine learning, empirical asset pricing, and backtest governance. Triple-barrier labeling and meta-labeling are widely associated with the financial machine learning framework popularized by Lopez de Prado. The motivation is sensible: financial outcomes are path dependent, and the timing of wins and losses matters. However, the existence of an event label does not automatically define an investable payoff.

The empirical asset-pricing literature similarly warns against treating statistical regularities as economically useful factors without careful testing. Factor discovery can be distorted by data mining, multiple testing, changing regimes, transaction costs, and implementation frictions. The present paper contributes a narrower but related point: even before one debates whether a factor is real, a target label may already be economically misaligned with the return object that a portfolio must realize.

The paper also follows the backtest-hygiene literature. A result is not treated as successful simply because one repaired a failed rule after observing the test period. Diagnostic repairs are informative, but they do not automatically produce a valid strategy. In this study, the best test-period diagnostic variant is not promoted because the family does not pass validation.

3. Data and Empirical Setting

The empirical setting is a proprietary U.S. equity / NASDAQ-derived research panel assembled for the FactorInvest research environment. Local lineage references ANUPY, JPY, and NASDAQ panel artifacts, but the paper-level description is a proprietary U.S. equity research panel.

The raw return panel contains 6,959,572 rows and 2,373 distinct tickers, with coverage from July 28, 1980 to March 7, 2025. The GC004 model-ready panels contain 3,552,232 rows and 2,282 distinct tickers, with coverage from January 26, 1981 to February 6, 2025. The primary weekly model-ready test sample contains 163,628 rows. The frozen signal test selection contains 863 distinct tickers, with approximately 11.89 selected names per weekly rebalance date on average.

The chronological split is: training observations dated on or before December 31, 2019; validation observations from January 1, 2020 to December 31, 2021; and test observations from January 1, 2022 onward. The primary weekly sample starts on January 3, 2022 and model data extend locally to February 6, 2025. The latest return-panel date observed in the local pack is March 7, 2025.

The underlying panel is daily. The primary modeling and candidate-selection policy is weekly_seq5, a weekly sampling cadence designed to reduce excessive overlap while preserving a practical weekly decision rhythm. The confirmatory lower-overlap policy is monthly_seq21, approximately aligned with the 21-trading-day target horizon.

3.1 Return Field and Benchmark

The paper backtest uses RETURN_ASSET, a daily arithmetic asset return field accepted by the local mark-to-market contract for paper backtesting. This field is not independently proven here to be a full production-grade total-return series. The analysis therefore treats it as a historical research return field, not as a complete investable return contract.

The primary benchmark is benchmark_high_liquidity_equal_weight. It is formed from observations with high_liquidity_flag equal to true on the signal date, uses RETURN_ASSET, and weights eligible names equally. The benchmark is aligned to the same entry dates and same 21 post-entry return days as the selected signal sleeve. It is intentionally an internal liquidity-aware research benchmark rather than an external investable index.

Table 1. Mark-to-market return contract used in the paper backtest.

Contract item	Status	Field	Coverage	Decision	Limitation
explicit mark to market return	accepted for paper backtest	RETURN_ASSET	99.95%	Use as daily arithmetic mark-to-market return for paper backtest only.	Corporate actions / total-return adjustment not yet proven.
price sanity field	available for alignment checks	PX_LAST	100.00%	Use for coverage sanity, not as model feature.	Backtest uses RETURN_ASSET, not reconstructed price returns.
benchmark reference return	available not primary	RETURN_BENCHMARK	99.97%	Keep as reference field; primary paper benchmark is equal-weight universe from RETURN_ASSET.	Benchmark construction details are inherited from JPY panel.
corporate actions total return	unresolved for production	nan		Do not promote to production-grade backtest until documented.	Paper backtest can run; production conclusion cannot.
return panel overall quality	diagnostic	RETURN_ASSET	96.45%	11 sector files loaded; date range 2019-12-23 to 2025-03-07.	0 duplicate DATA/TICKER rows removed.

Note. RETURN_ASSET is accepted for paper backtesting only. Corporate actions and total-return production treatment remain unresolved.

4. Triple-Barrier Target Design and Model

For each asset i at decision date t , the primary triple-barrier target evaluates the future path over $H = 21$ trading days. The upper barrier is a +10% take-profit threshold, the lower barrier is a -5% stop-loss threshold, and the vertical barrier occurs at $t + H$. The label equals +1 if the upper barrier is touched first, -1 if the lower barrier is touched first, and 0 if neither is touched before the horizon expires.

The primary target is TBM_LONG_FIX_21D_SL5_TP10. Its interpretation is: long-side triple-barrier model, fixed 21-trading-day horizon, 5% stop-loss, and 10% take-profit. A longer-horizon target, TBM_LONG_FIX_63D_SL10_TP20, is used as a related sensitivity target.

The model producing p_take_profit and p_stop is the GC004 score-only baseline, identified in the local manifest as `weighted_linear_ridge_multiclass_closed_form`. It is a weighted multiclass linear ridge classifier trained on approved composite factor scores. Its outputs are class probabilities for the triple-barrier label states. The model is intentionally treated as an interpretable baseline rather than a production timing engine.

The approved feature groups include price-action recovery, defensive compounds, value and quality, cash profitability, Piotroski-style value and health, distress avoidance, earnings red flags, QARP / recovery composites, and defensive value composites. Identifiers, context fields, targets, labels, sampling flags, and forward returns are separated from approved model features. TBM labels and future returns are not used as feature inputs.

- Training sample: observations dated on or before 2019-12-31.
- Validation sample: 2020-01-01 to 2021-12-31.
- Test sample: 2022-01-01 onward, with the primary weekly test beginning on 2022-01-03.
- Primary policy: `weekly_seq5`; confirmatory lower-overlap policy: `monthly_seq21`.

5. From Label Prediction to Portfolio Validation

The empirical design follows a ladder from intermediate classification evidence to economic validation. The first layer asks whether the model improves the probability ranking of the take-profit label. The second layer asks whether the same ranking improves event returns. The third layer converts the frozen rule into a temporal-safe portfolio construction. The fourth layer evaluates the selected sleeve against a high-liquidity equal-weight benchmark after round-trip costs. The fifth layer records repair attempts and applies validation discipline.

The candidate rule gates eligible names and then ranks them by p_take_profit within each signal date. It selects the top 10% of eligible names per rebalance date and weights selected names equally. The selected names are held over a 21-trading-day forward window. The base transaction-cost assumption is 50 bps round-trip, with sensitivities at 0, 10, 25, 50, and 100 bps.

6. Empirical Results

6.1 Label Lift Does Not Imply Return Lift

The primary target shows modest but real label discrimination. Under the `weekly_seq5` policy, the test AUC for take-profit versus rest is 0.580. The base take-profit share is 25.70%, while the top decile ranked by p_take_profit has a take-profit share of 30.71%. This is a 5.01 percentage-point lift.

However, the same top decile has a negative event-return lift of -0.67 percentage points and a stop-label share of 67.26%. The result is not that the classifier learned nothing. It learned something about the label. The problem is that the label was not aligned with the economic payoff the portfolio needed.

Table 2. Label lift versus event-return lift for fixed long triple-barrier targets.

Target	Policy	Rows	AUC TP vs rest	Base TP share	Top decile TP share	TP lift	Event-return lift	Top decile stop share
TBM LONG FIX 21D SL5 TP10	monthly_seq21	39,116	0.582	25.67%	30.67%	+5.00 pp	-0.84 pp	67.33%
TBM LONG FIX 21D SL5 TP10	weekly_seq5	163,628	0.580	25.70%	30.71%	+5.01 pp	-0.67 pp	67.26%
TBM LONG FIX 63D SL10 TP20	monthly_seq21	37,441	0.573	24.43%	28.30%	+3.87 pp	-1.53 pp	69.72%
TBM LONG FIX 63D SL10 TP20	weekly_seq5	156,628	0.574	24.53%	28.03%	+3.50 pp	-1.60 pp	70.17%

Note. TP lift is measured as top-decile take-profit share minus base take-profit share. Event-return lift is measured as top-decile event return minus base event return.

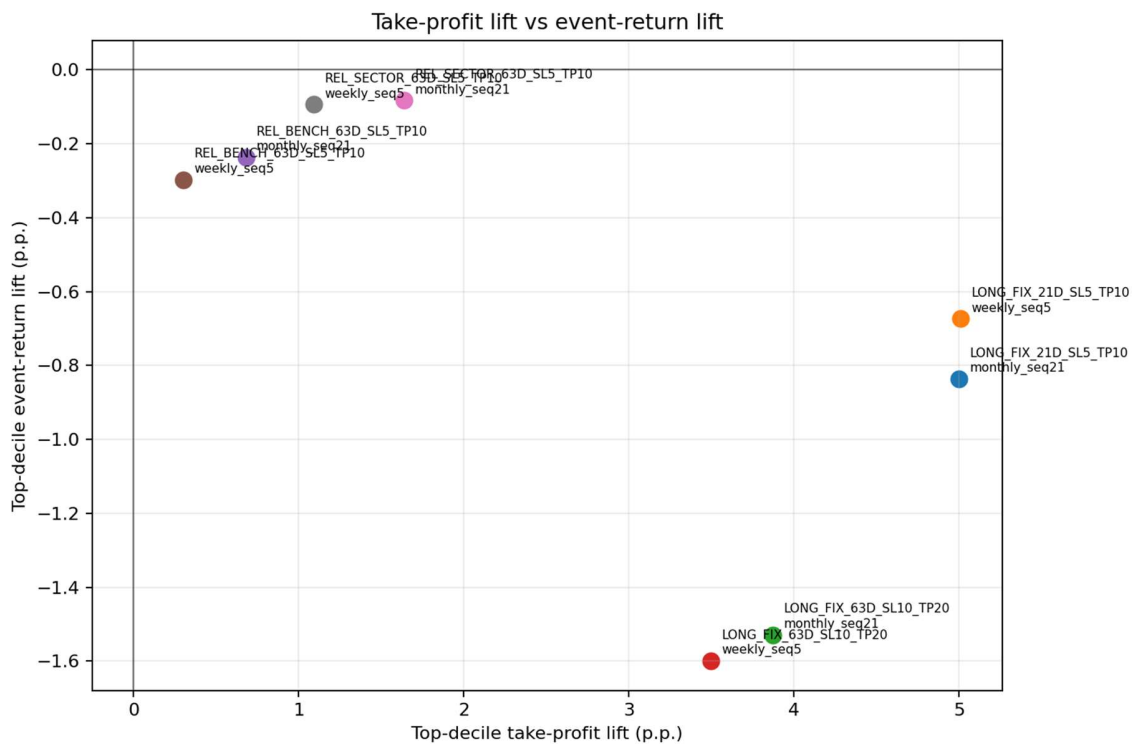


Figure 1. Label lift does not imply return lift. Positive take-profit label lift coexists with negative event-return lift across target and policy settings.

6.2 Probability Decile Mismatch

The full prediction-universe diagnostic makes the mismatch more visible. Sorting by predicted take-profit probability increases the realized take-profit share from 13.54% in the bottom decile to 30.71% in the top decile. Yet the same top decile has a stop share of 67.26% and a mean high-liquidity benchmark-relative excess return of -3.59% over the 21-day horizon.

This is the core failure mode. The top decile is not merely noisy. It has the property the model was trained to find, but it also has poor economic behavior. Probability of hitting a take-profit barrier is not equivalent to expected portfolio payoff.

Table 3. Decile diagnostic: ranking by predicted take-profit probability.

Decile	Rows	TP share	Stop share	Mean event return	Mean excess vs high-liq 21D
1	16,362	13.54%	33.35%	0.11%	-0.17%
2	16,363	19.01%	42.65%	0.09%	-0.13%

Decile	Rows	TP share	Stop share	Mean event return	Mean excess vs high-liq 21D
3	16,363	22.36%	46.69%	0.24%	0.23%
4	16,363	23.85%	51.25%	-0.11%	-0.04%
5	16,363	27.12%	53.76%	-0.05%	-0.04%
6	16,362	28.51%	57.33%	-0.20%	-0.50%
7	16,363	30.44%	60.05%	-0.24%	-0.52%
8	16,363	30.06%	63.42%	-0.64%	-1.10%
9	16,363	31.40%	64.19%	-0.46%	-1.15%
10	16,363	30.71%	67.26%	-0.89%	-3.59%

Note. Rows refer to the cycle014 full prediction-universe test split. Excess is measured relative to the internal high-liquidity equal-weight benchmark over the 21-day horizon.

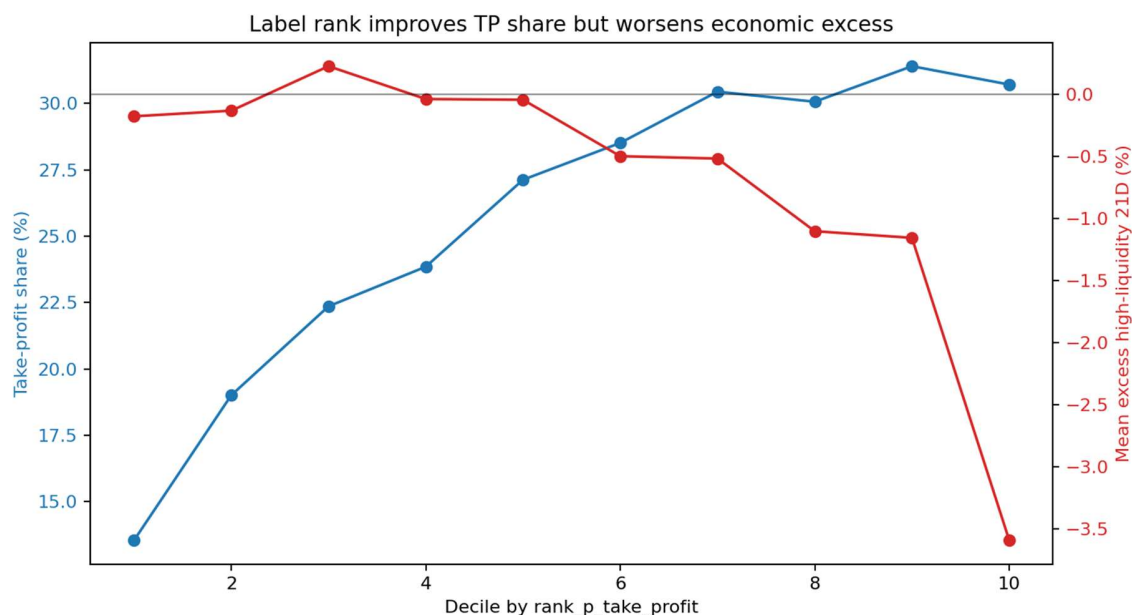


Figure 2. Probability decile mismatch. Higher predicted take-profit probability raises take-profit share but worsens benchmark-relative economic excess in the upper deciles.

6.3 Mark-to-Market Portfolio Backtest

The frozen 21-day weekly rule is then evaluated as a mark-to-market portfolio. At the 50 bps round-trip base cost, the selected portfolio returns -37.90% during the test window, while the high-liquidity equal-weight benchmark returns -13.83%. The resulting excess total return is -30.59%, and the excess CAGR is -10.51 percentage points.

The cost sensitivity shows that the failure is not merely a transaction-cost story. Even at zero cost, the portfolio underperforms the benchmark by -15.45% total return. Increasing costs makes the result worse, but the underlying benchmark gap exists before costs.

Table 4. Mark-to-market performance by round-trip transaction cost.

Cost bps	Portfolio total	Benchmark total	Excess total	Portfolio CAGR	Benchmark CAGR	Excess CAGR	Max DD	Positive excess day share
0	-24.34%	-13.83%	-15.45%	-8.13%	-4.42%	-4.97%	-39.25%	45.60%
10	-27.27%	-13.83%	-18.72%	-9.23%	-4.42%	-6.11%	-40.65%	45.24%
25	-31.46%	-13.83%	-23.39%	-10.85%	-4.42%	-7.78%	-42.68%	44.87%
50	-37.90%	-13.83%	-30.59%	-13.48%	-4.42%	-10.51%	-45.92%	44.03%
100	-49.04%	-13.83%	-43.03%	-18.53%	-4.42%	-15.72%	-51.87%	41.98%

Note. The benchmark is the internal high-liquidity equal-weight benchmark. Cost is applied as round-trip bps.

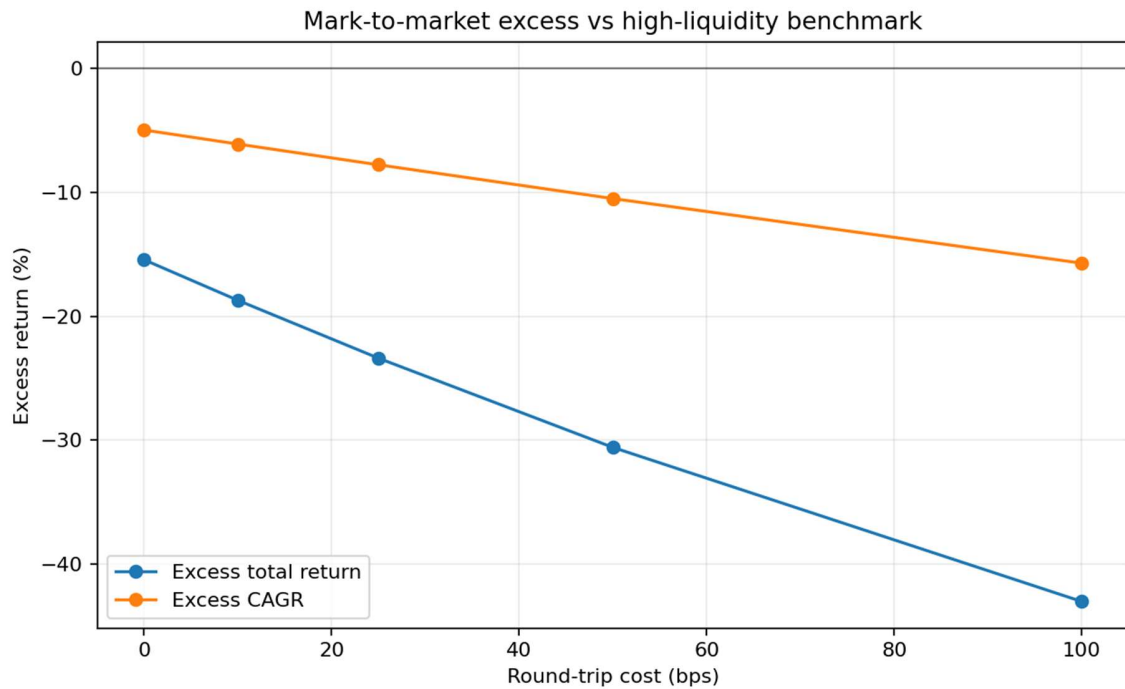


Figure 3. Mark-to-market excess return by round-trip cost. The selected sleeve remains below the benchmark even at zero cost.

7. Failure Mode Diagnostics

The diagnostics identify five main failure modes. The benchmark gap is not only cost; the strategy is already behind at zero bps. The selected sleeve also has material healthcare concentration. Healthcare represents 51.83% of selected exposure on average in the test split versus 34.78% in the high-liquidity benchmark. A high-liquidity-only repair does not solve the problem, which implies that liquidity improves implementability but does not create alpha. Most importantly, probability ranking itself is mismatched: a higher p_take_profit is not a sufficient expected-payoff ranking.

Table 5. Main failure modes.

Failure mode	Evidence	Severity	Repair implication
benchmark gap not only cost	0 bps excess vs high-liquidity = -15.45%; 50 bps = -30.59%	high	Nao basta reduzir custo; o ranking precisa bater high-liquidity bruto.
healthcare concentration	Healthcare share selected=51.83%; high-liq benchmark=34.78%; non-healthcare prototype excess=-9.10%	high	Testar ranking sector-neutral ou limite setorial com reposicao, nao apenas exclusao.
liquidity filter not sufficient	high_liquidity_only prototype excess=-34.88%	medium	Liquidez melhora investibilidade, mas nao conserta alpha.
probability ranking mismatch	p_margin_p50 prototype excess=-33.16%; p_take_profit top-minus-bottom mean excess=-0.61%	high	Probabilidade de take-profit nao ordena retorno mark-to-market; revisar target/ranking composto.
production return contract still open	RETURN_ASSET aceito para paper; corporate actions/total return segue unresolved for production.	medium	Antes de producao, documentar total return e eventos corporativos.

Note. The failure modes are interpreted as research diagnostics, not as evidence that all triple-barrier labels fail.

Table 6. Sector concentration in the selected test sleeve.

Sector	Selected share	Benchmark share	Mean fwd 21D	Mean excess 21D	Share gap	Overrep. ratio
Healthcare	51.83%	34.78%	-0.79%	-1.01%	17.05%	1.49x
Technology	18.05%	16.26%	-0.03%	-0.40%	1.79%	1.11x
Consumer Cyclical	9.12%	8.94%	0.57%	0.08%	0.18%	1.02x
Industrials	6.52%	8.87%	0.70%	0.25%	-2.35%	0.74x
Communication Services	4.45%	5.61%	-0.70%	-0.30%	-1.16%	0.79x
Financial Services	3.41%	15.26%	1.37%	1.51%	-11.85%	0.22x
Consumer Defensive	2.81%	3.90%	-0.92%	-1.06%	-1.09%	0.72x

Sector	Selected share	Benchmark share	Mean fwd 21D	Mean excess 21D	Share gap	Overrep. ratio
Basic Materials	1.71%	2.07%	1.10%	0.54%	-0.36%	0.83x

Note. Selected share and benchmark share are average date-level shares in the test split. Healthcare concentration is a major diagnostic, but not the only source of failure.

8. Repair Attempts and Validation Discipline

The research path does not stop at the first negative backtest. Several pragmatic repairs are tested: healthcare exclusion, high-liquidity filters, stop-probability gates, sector caps, and ranking reformulations. Some reduce the damage. The non-healthcare diagnostic, for example, improves excess total return from -30.59% to -9.10% at the 50 bps setting. However, it still does not beat the high-liquidity benchmark.

The most tempting result is the cap25_defensive_value diagnostic, which almost matches the benchmark in the test split. It is not promoted because the family does not pass the validation lock. This is an important governance point. Once the test period has been inspected, a good-looking test-only repair is diagnostic evidence, not a validated strategy.

Table 7. Repair ladder and validation outcomes.

Stage	Variant	Split	Cost bps	Portfolio total	Benchmark total	Excess total	Decision
cycle011 first mtm backtest	21d_weekly_frozen_rule	test	50	-37.90%	-13.83%	-30.59%	do not promote after first paper backtest
cycle012 simple repair	non_healthcare	test	50	-17.24%	-13.89%	-9.10%	diagnostic only
cycle012 simple repair	highliq_p_stop_p40	test	50	-30.17%	-13.90%	-21.32%	diagnostic only
cycle012 simple repair	p_stop_p40	test	50	-29.42%	-13.75%	-22.13%	diagnostic only
cycle013 sector neutral rebuild	sector_cap_25	validation	50	33.42%	60.89%	-18.42%	no repaired rule locked
cycle013 sector neutral rebuild	sector_cap_25	test	50	-29.57%	-13.83%	-22.03%	no repaired rule locked
cycle014 target ranking reformulation	cap25_p_take_profit	validation	50	33.42%	60.89%	-18.42%	no ranking rule locked
cycle014 target ranking reformulation	cap25_p_take_profit	test	50	-29.57%	-13.83%	-22.03%	no ranking rule locked
cycle014 best test diagnostic not locked	cap25_defensive_value	test	50	-4.50%	-13.83%	-0.14%	diagnostic only not locked

Note. Diagnostic variants that reduce damage are not promoted unless the rule family passes validation before test interpretation.

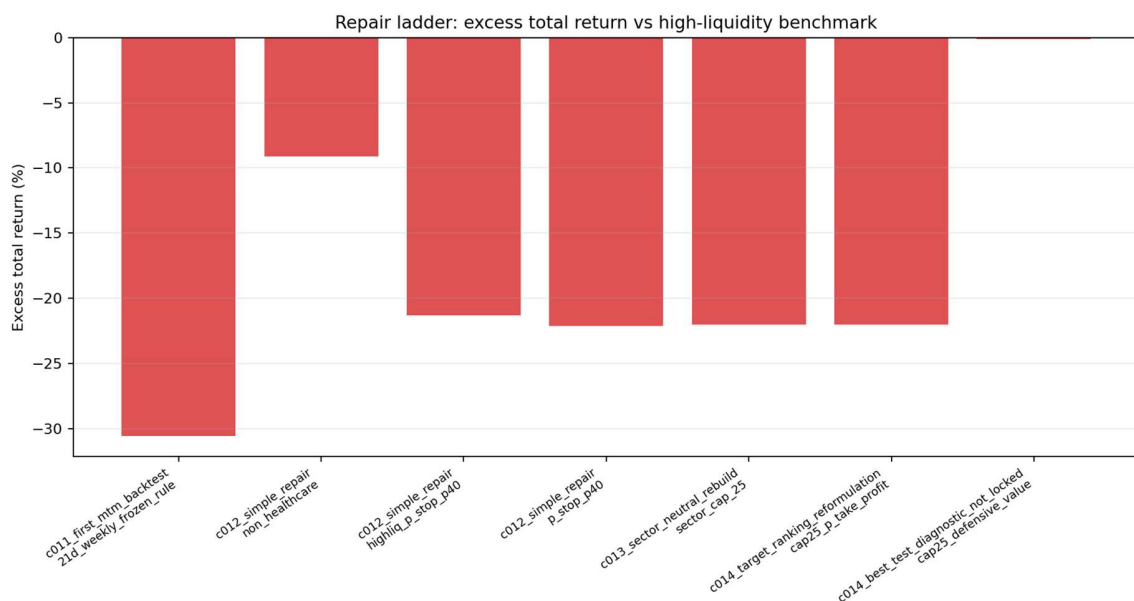


Figure 4. Repair ladder excess return. Several repairs reduce damage, but no rule is locked as validated.

9. Discussion

The empirical lesson is not that triple-barrier labeling is useless. The lesson is narrower and more useful: a target label is an intermediate representation, not the final economic object. The more economically complex the desired outcome, the more dangerous it becomes to rely on label metrics alone.

In this case, the model increases the probability of selecting take-profit events, but the selected group also carries high stop incidence, poor benchmark-relative return, and sector concentration. The label asks whether a price path touches a barrier. The portfolio asks whether a capital allocation beats a benchmark after costs, risk, concentration, timing, and implementation constraints. These are not the same question.

This suggests a practical research agenda. Target engineering should move closer to expected economic payoff. Ranking functions should incorporate payoff magnitude, stop probability, benchmark-relative return, concentration constraints, and portfolio construction. Validation should move from classification lift to return lift, then to benchmark lift, then to mark-to-market portfolio curves.

10. Limitations

This paper is a historical empirical case study, not a production strategy. The return field used in the mark-to-market backtest is accepted by the local artifacts for paper backtesting, but it is not independently proven here to be a fully adjusted total-return series. Dividends, corporate actions, delisting treatment, borrow mechanics, slippage, taxes, market impact, capacity, and other implementation frictions are not fully modeled.

The universe is a proprietary U.S. equity research panel. Survivorship, listing-history completeness, point-in-time universe construction, and delisting treatment are not independently proven in this pack. These issues can materially affect factor-investing evidence and should be resolved before any production interpretation.

The paper also documents a specific failure mode in one family of triple-barrier labels and one research environment. It is not a general rejection of triple-barrier labeling, event-based labeling, meta-labeling, or machine-learning classification in finance. The appropriate claim is that label success should not be treated as portfolio-return success without an explicit economic bridge.

11. Conclusion

The GC004 research path found a predictable triple-barrier label, froze a temporal-safe rule, and tested it as a mark-to-market portfolio. That sequence is valuable because it revealed where the signal failed. The top predicted take-profit decile improved take-profit frequency but did not improve economic returns. The frozen portfolio lost to the high-liquidity equal-weight benchmark, even before transaction costs. Repair attempts did not produce a validation-locked rule.

The conclusion is therefore negative for strategy promotion but positive as methodology. Target labels are not portfolio returns. Before a label-driven strategy is treated as investable, the label should be bridged to a return contract that includes benchmark comparison, cost, risk, liquidity, concentration, temporal safety, and validation discipline.

The model learned the label. It did not learn the trade.

Appendix A. Practical Validation Checklist

Table A1. Practical validation checklist for label-driven strategies.

Check	Question
Label lift	Does the model improve the target event relative to the base rate?
Return lift	Does the same ranking improve event return, not just event frequency?
Benchmark lift	Does the signal improve return relative to a relevant benchmark?
Stop asymmetry	Does the top decile also increase stop probability or loss severity?
Sector concentration	Is the signal simply loading into one sector?
Liquidity	Does the result survive high-liquidity filters?
Temporal safety	Are signals generated only with information available at the decision date?
Mark-to-market	Does the frozen rule survive as a portfolio curve after costs?
Validation lock	Does the rule pass validation before test interpretation?
Return contract	Is the return field production-valid, including corporate actions and total-return treatment?

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